

Blind Disturbance-Model Selection for Diffusion-Aided Semantic Image Reception

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Abstract—We present a blind disturbance-model selector for diffusion-aided semantic image reception. Given a received latent, it selects Gaussian denoising or joint signal–interference reconstruction and decodes the recovered latent into an image. Rather than reconstructing an unknown interferer on every frame, the selector uses received energy and block-energy heterogeneity to choose one receiver before diffusion inference. The rule requires only $O(n)$ calculations and no neural retraining; its thresholds are fitted on development images and frozen before evaluation. Experiments use CelebA images with CIFAR-10 interference and identical observations across receivers. On 256 independent image pairs with two random seeds and 18 channel conditions, the frozen rule reaches 21.593 dB mean PSNR. It improves on calibrated joint sampling by +1.334 dB and on blind Gaussian denoising by +0.750 dB. The rule uses 52.2 predictor evaluations per frame on average, compared with 80 for joint sampling. An additional same-gate control tests whether the gain extends beyond bypassing joint inference; separately trained DeepJSCC and MambaJSCC codecs provide external references under the same channel-use budget.

Index Terms—Semantic communications, interference cancellation, diffusion models, receiver adaptation, deep joint source–channel coding.

I. INTRODUCTION

Semantic image communication requires recovering image content from limited, corrupted channel observations. Deep joint source–channel coding (JSCC) addresses this task by mapping images directly to channel symbols, with gradual quality degradation as noise increases [1]. SwinJSCC and MambaJSCC extend this approach through transformer and state-space architectures [2], [3]. Generative receivers further exploit learned priors to recover corrupted representations. Even so, co-channel interference of unknown power and structure leaves a deployed receiver with a choice: approximate the disturbance as noise or reconstruct the interfering signal explicitly.

CDDM and ICDM provide two relevant recovery mechanisms. CDDM matches the aggregate disturbance to a diffusion noise level [4], whereas ICDM jointly reconstructs desired and interfering latents [5]. Iterative source error correction [6], channel-conditioned diffusion reconstruction [7], and blind inverse-problem inference [8] also advance receiver adaptation. However, joint reconstruction must infer two latents and unknown interference amplitudes from one mixture. Estimation errors can offset the benefit of modeling interference structure while consuming additional computation. Gaussian denoising avoids that estimation burden but discards useful structure. This raises the question: can the receiver choose

between these models from the observation alone, before paying for generative inference?

In this paper, we take a step toward observation-dependent generative reception. We introduce a blind disturbance-model selector that takes a received latent and chooses either Gaussian denoising or joint reconstruction. The selected receiver recovers the desired latent for a common image decoder. The selector computes two energy statistics in $O(n)$ time and executes only one branch. All neural weights remain frozen. This design tests whether changing the receiver decision alone can improve image recovery, without changing the transmitted representation or training new neural modules.

Our key observation is that explicit interference modeling offers structure at the cost of estimation uncertainty. Its value can therefore vary across received frames. To inform this choice, we use excess received energy to measure aggregate disturbance and block-energy heterogeneity to describe its variation across latent blocks. Neither statistic identifies the interfering image; desired-signal variation also affects both. We therefore use image-group validation to compare energy-only and two-feature rules on separate development images, retaining a fixed receiver if adaptation gives no validation gain. The chosen rule is frozen before independent confirmation.

Experiments on 256 independent image pairs show a mean PSNR gain of 0.750 dB over blind Gaussian denoising and 1.334 dB over calibrated joint sampling, with 34.7% fewer predictor evaluations than joint sampling. These gains require no neural retraining. A same-gate control tests whether selection adds value beyond bypassing joint inference. Profile-wise comparisons reveal that the preferred fixed receiver changes with interference structure, while harmful decisions and block mismatch delimit the benefit. Separately trained DeepJSCC and MambaJSCC provide external codec references. Our shared-prior receiver ports do not reproduce the complete published CDDM or ICDM systems.

Our contributions are threefold. (1) We introduce a blind selector that chooses a disturbance model before generative inference using two energy statistics. (2) We show that the preferred fixed receiver differs across interference profiles, motivating observation-dependent selection. (3) We establish a mean-quality and computation benefit on independent paired observations, while quantifying harmful decisions and mismatch sensitivity. Sections II and III present the system model and selector; Sections IV and V report experiments and conclusions.

II. SYSTEM MODEL AND PROBLEM FORMULATION

In this section, we define the communication chain and receiver information, then formulate model selection under fixed bandwidth, power, and weights.

A. Image Representation and Transmission

Let $\mathbf{u} \in [0, 1]^d$ be an RGB image, with $d = 3H_{\text{img}}W_{\text{img}}$. The JSCC encoder f_s combines representation learning and channel mapping; f_z encodes an interfering image $\mathbf{v}$ into the same dimension:

$$\mathbf{x} = \mathcal{N}(f_s(\mathbf{u})), \quad \mathbf{z} = \mathcal{N}(f_z(\mathbf{v})), \quad \mathcal{N}(\mathbf{t}) = \frac{\sqrt{n}\mathbf{t}}{\|\mathbf{t}\|}. \quad (1)$$

Each nonzero codeword has unit average power and bandwidth ratio n/d . For B blocks of lengths L_b , with $\sum_b L_b = n$, the received symbols are

$$\mathbf{y}_b = \mathbf{x}_b + a_b \mathbf{z}_b + \mathbf{w}_b, \quad \mathbf{w} \sim \mathcal{CN}(\mathbf{0}, N_0 \mathbf{I}), \quad a_b \geq 0. \quad (2)$$

Here a_b is the unknown interference amplitude and N_0 the complex noise power. Desired-channel gain is unity; symbol alignment is known. The receiver knows N_0 , the normalization, and block boundaries, but not $\mathbf{u}$, $\mathbf{z}$, or a_b . For nominal SINR γ , the simulator sets the average interference power to $10^{-\gamma/10} - N_0$. Realized block energies of $\mathbf{z}$ make the per-frame SINR differ from this nominal setting.

B. Diffusion-Assisted Image Recovery

The receiver combines the frame-specific evidence in $\mathbf{y}$ with learned latent priors. Joint reconstruction uses both priors:

$$p(\mathbf{x}, \mathbf{z} \mid \mathbf{y}, \mathbf{a}) \propto p_s(\mathbf{x})p_z(\mathbf{z}) \times \exp \left[-\frac{1}{N_0} \sum_b \|\mathbf{y}_b - \mathbf{x}_b - a_b \mathbf{z}_b\|^2 \right]. \quad (3)$$

The amplitudes must be estimated. Moreover, a matching mixture cannot identify both latents: $(\mathbf{x}_b + a_b \delta_b, \mathbf{z}_b - \delta_b)$ produces the same observation. Gaussian recovery instead uses only p_s :

$$p_G(\mathbf{y} \mid \mathbf{x}; \nu) = \mathcal{CN}(\mathbf{y}; \mathbf{x}, \nu \mathbf{I}), \quad \nu \geq N_0. \quad (4)$$

The Gaussian and joint receivers, $\mathcal{R}_G$ and $\mathcal{R}_S$, use frozen diffusion priors with observation-based initialization or guidance. Neither performs exact posterior sampling, and only joint recovery estimates a second latent. A blind rule $\pi_\theta(\mathbf{y}) \in \{\mathcal{G}, \mathcal{S}\}$ selects one branch. The common decoder then reconstructs

$$\hat{\mathbf{u}}_\theta = g_s(\mathcal{N}(\mathcal{R}_{\pi_\theta(\mathbf{y})}(\mathbf{y}))). \quad (5)$$

C. Quality, Cost, and Receiver Selection

We measure image reconstruction with per-image PSNR, $q(\mathbf{u}, \hat{\mathbf{u}}) = -10 \log_{10}(\|\mathbf{u} - \hat{\mathbf{u}}\|^2/d)$ for clipped RGB outputs. LPIPS and MS-SSIM measure fidelity, not downstream accuracy. Computation counts calls to both priors; receiver latency includes selection and decoding.

For development pairs $\mathcal{D} = \{(\mathbf{u}_i, \mathbf{y}_i)\}_{i=1}^M$, we fit

$$\theta^* \in \arg \max_{\theta \in \Theta} \frac{1}{M} \sum_{i=1}^M q(\mathbf{u}_i, \hat{\mathbf{u}}_\theta(\mathbf{y}_i)), \quad (6)$$

where Θ contains the rule family retained by image-group validation and a constant-receiver fallback. Bandwidth, power, and neural weights remain fixed; computation is reported separately. Fitting is offline; at inference, the frozen rule executes one branch without knowing its reconstruction error.

III. BLIND DISTURBANCE-MODEL SELECTION

In this section, we present the candidate receivers, energy statistics, and rule-fitting procedure that map received symbols to a receiver choice.

A. Candidate Receivers

Both receivers share the frozen desired-signal prior and decoder, isolating the disturbance-model choice. To account for unknown interference power, the Gaussian receiver estimates aggregate disturbance from the received symbols:

$$e_b = \frac{\|\mathbf{y}_b\|^2}{L_b}, \quad \bar{e} = \frac{1}{n} \sum_b L_b e_b, \quad (7)$$

$$\hat{P} = [\bar{e} - 1 - N_0]_+, \quad \hat{\nu} = N_0 + \hat{P}.$$

With unit source power and a vanishing signal–interference cross expectation, unclipped excess energy estimates the mean interference energy. Finite blocks, latent correlation, and clipping make this estimate inexact. In particular, local desired-signal energy can resemble interference heterogeneity.

Let $r(\mathbf{y}) \in \mathbb{R}^{2n}$ stack real and imaginary components. Given the training diffusion schedule $\bar{\alpha}_t$, the Gaussian receiver G starts at the nearest noise level satisfying

$$\frac{1 - \bar{\alpha}_{t_0}}{\bar{\alpha}_{t_0}} \simeq \frac{\hat{\nu}}{2}, \quad \mathbf{v}_{t_0} = \sqrt{\bar{\alpha}_{t_0}} r(\mathbf{y}). \quad (8)$$

The factor 1/2 converts complex noise power to variance per real coordinate. For decreasing respaced indices, a noise predictor ϵ_s gives

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{v}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_s(\mathbf{v}_t, t)}{\sqrt{\bar{\alpha}_t}}, \quad (9)$$

$$\mathbf{v}_{t'} = \sqrt{\bar{\alpha}_{t'}} \hat{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_{t'}} \epsilon_s(\mathbf{v}_t, t). \quad (10)$$

The receiver uses at most 40 predictor evaluations and fewer when the matched initial index is below 39.

To exploit interference structure, the joint receiver S initializes $a_b^{(0)} = \sqrt{[e_b - 1 - N_0]_+}$ and performs joint diffusion inference with signal and interference predictors. During late corrector updates it obtains denoised, frame-normalized estimates $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{z}}$, then updates

$$a_b^+ = \left[\frac{\Re\{\tilde{\mathbf{z}}_b^H(\mathbf{y}_b - \tilde{\mathbf{x}}_b)\}}{\|\tilde{\mathbf{z}}_b\|^2} \right]_+. \quad (11)$$

An empty denominator retains the previous amplitude. This nonnegative least-squares update is conditional on point estimates and does not correct their uncertainty. Calibration begins when the signal coefficient $\alpha(t) \geq 0.5$. Estimated block SINR updates the inherited guidance coefficients by clipped linear interpolation of the original AWGN table. The existing 40-step, order-three sampler calls each prior 40 times. An initialization-only variant keeps the amplitudes fixed and isolates the effect of calibration.

B. Selection Before Generative Inference

To distinguish frames with similar total energy but different block profiles, the selector combines $\hat{P}$ with normalized block-energy variation:

$$H = \frac{B^{-1} \sum_{b=1}^B (e_b - \bar{e})^2}{\max(\bar{e}^2, \varepsilon)}. \quad (12)$$

Here $\varepsilon > 0$ prevents division by zero, and all receiver blocks have equal length. The statistic H measures energy variation, not the interference distribution. For $\theta = (\tau_\ell, \tau_h, \eta)$, the rule is

$$\pi_\theta(\mathbf{y}) = \begin{cases} \text{G}, & \hat{P} \leq \tau_\ell, H \leq \eta, \\ \text{G}, & \hat{P} \leq \tau_h, H > \eta, \\ \text{S}, & \text{otherwise.} \end{cases} \quad (13)$$

We also test the reversed assignment of G and S in (13); thus, the Gaussian branch is not assigned to the low-power region a priori. The energy-only family sets $\tau_\ell = \tau_h$. Either family can reduce to a constant receiver, allowing validation to reject adaptation.

To assess transfer across images, eight development folds keep each image's conditions together. We fit thresholds on seven folds and evaluate PSNR on the eighth, then refit the best out-of-fold family on all development images. Ties favor lower cost, then the simpler family. If neither family improves on the better fixed receiver, validation retains that receiver. Confirmation and pressure results never alter the thresholds. At inference, the selector evaluates (13) and executes only the chosen branch. This adds $O(n)$ energy calculations and no diffusion evaluations.

For analysis, let $q_j(\mathbf{u}, \mathbf{y})$ be a candidate's PSNR. The unavailable per-frame oracle has quality $\max(q_G, q_S)$. The exact identity

$$\max(q_G, q_S) - q_{\pi_\theta} = |q_G - q_S| \mathbf{1}\{\pi_\theta \neq \arg \max_j q_j\} \quad (14)$$

shows that complementary receivers alone are insufficient: errors on large-gap frames dominate regret. This identity is an accounting relation, not a generalization bound. The fitting objective is mean PSNR; it does not optimize mean MSE, LPIPS, or harmful-tail probability.

IV. EXPERIMENTAL EVALUATION

In this section, we test whether selecting one frozen receiver improves image quality while reducing computation. We compare fixed receivers and external codecs, then examine harmful decisions and interference mismatch.

A. Experimental Setup

We use 128×128 CelebA center crops as desired images [9]; 32×32 CIFAR-10 interference images are nearest-neighbor upsampled. Training uses 162,770 desired and 50,000 interference images. The frozen SwinJSCC codec has stage widths (128, 192, 256), depths (2, 2, 6), and an $8 \times 16 \times 16$ real latent: 1,024 complex symbols at CBR 1/48. Each DiT prior has 28 blocks, width 1,152, 16 heads and patch size two. The

inherited training schedule uses 40/100 desired/interference-codec epochs and 50/20 desired/interference-prior epochs; selection adds no neural training.

DeepJSCC and MambaJSCC each train desired and interference codecs for 40 epochs at SNR 20 dB and CBR 1/48, with one seed and validation-MSE checkpoint selection. DeepJSCC uses a pinned third-party PyTorch reproduction; MambaJSCC uses official code. Public checkpoints also provide transfer diagnostics without fine-tuning, using one encoder for both streams. Public MambaJSCC differs in training domain, channel and resolution; public DeepJSCC also uses CBR 0.16256. Either codec would require newly trained, compatible diffusion priors. Complete published CDDM/ICDM systems are not reproduced.

Development uses 64 validation pairs and seed 1700. For independent confirmation, we use 256 test pairs, seeds 2700/2701, SNR 20 dB, and nominal SINRs $\{-7, -4, 0, 4, 7, 10\}$ dB. Sixteen 64-symbol blocks have stationary, alternating (0.25, 1.75), or burst (0, 2) relative interference powers. These 18 conditions give 9,216 frames per receiver. We run confirmation only after freezing the policy. File hashes exclude overlap with development, pressure tests, and 200 previously evaluated pairs; independence is at image-pair, not identity, level. The power grid is $\{0, 0.1, 0.2, 0.35, 0.5, 0.75, 1, 1.5, 2, 3, 4, 8, 10^6\}$; the heterogeneity grid is $\{0.05, 0.1, 0.2, 0.4, 0.8, 1.6\}$, giving 26 energy-only and 2,028 two-feature rules across both assignment directions.

We compare with direct decoding, Gaussian reception using N_0 alone, blind Gaussian reception, initialization-only and calibrated joint sampling, and the selector. The same-gate control uses direct decoding for $\hat{P} \leq 0.15$ and calibrated joint sampling otherwise. Its outputs and component times are reconstructed from paired branches, excluding gate overhead. Development also tests coupled feedback and a Gaussian receiver given nominal disturbance power, a privileged diagnostic. All SwinJSCC variants share received tensors and sampler seeds; external codecs share image pairs, conditions and channel seeds but generate their own codewords.

Measurements use float32, batch size one, TF32 disabled, four CPU threads, and an RTX 4090. Synchronized receiver time includes selection, sampling and decoding, excluding encoding, metrics and loading. Function evaluations (NFE) count calls to both priors. Quality metrics are per-image PSNR and MSE, LPIPS-VGG v0.1 [10], and four-scale MS-SSIM [11], averaged arithmetically after RGB clipping to $[0, 1]$; four scales fit the image resolution. Paired 95% intervals use 5,000 bootstrap resamples of image pairs, retaining both seeds and all conditions. A harmful decision loses more than 0.5 dB against its comparator.

B. Frozen Rule and Receiver Comparison

We first test whether block heterogeneity improves rule selection under the same image-group folds. Energy-only selection yields 22.002 dB, whereas the two-feature family

TABLE I
RECONSTRUCTION QUALITY AND RECEIVER COST ON THE
CONFIRMATION SET.

Receiver	PSNR	MSE $\times 10^3$	LPIPS	MS-SSIM	NFE	ms
SwinJSCC direct	18.67	29.14	0.440	0.697	0.0	8
DeepJSCC direct	17.39	24.71	0.575	0.684	0.0	1
MambaJSCC direct	16.92	44.13	0.463	0.613	0.0	10
Gaussian (N0)	18.83	28.70	0.438	0.700	18.0	225
Blind Gaussian	20.84	24.90	0.378	0.782	40.0	488
Init. joint	19.61	20.15	0.363	0.793	80.0	1110
Calibrated joint	20.26	18.49	0.362	0.802	80.0	1180
Joint + gate	20.45	18.43	0.361	0.801	65.9	974
Selector	21.59	16.45	0.336	0.832	52.2	701

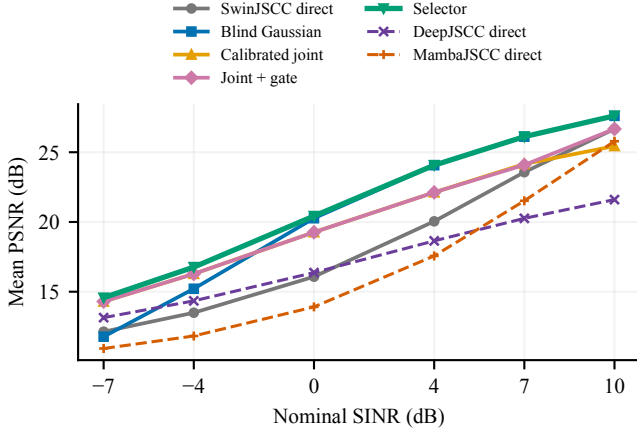


Fig. 1. Mean PSNR versus nominal SINR over the three interference profiles.

yields 22.239 dB. We therefore retain the energy and heterogeneity rule with $\tau_\ell = 8$, $\tau_h = 0.5$ and $\eta = 0.2$, with Gaussian reception below the applicable power threshold.

The low-heterogeneity development region contains 810 frames and has a maximum excess-energy estimate of 5.241. The fitted threshold 8 therefore lies beyond observed support and makes this region always Gaussian. This threshold is not an identified physical transition, and the family comparison does not establish that either feature is individually necessary. The earlier coupled-feedback receiver obtains 20.477 dB, versus 20.605 dB for calibrated joint sampling and 20.835 dB for its same-gate control. The privileged nominal-power Gaussian diagnostic obtains 21.591 dB. These development comparisons motivate the receiver decision but are not independent confirmation. We next test the frozen rule on independent paired observations (Table I and Fig. 1). The selector changes mean PSNR by +1.334 dB relative to calibrated joint sampling (95% interval [1.265, 1.404]), +0.750 dB relative to blind Gaussian denoising (95% interval [0.698, 0.806]), and +1.140 dB relative to the same-gate control (95% interval [1.075, 1.205]). Both fixed-receiver comparisons have positive paired intervals. The same-gate comparison also supports a benefit beyond the tested bypass alone. These results establish a mean-PSNR gain within the matched population. Estimating total disturbance changes the Gaussian receiver from 18.829 to

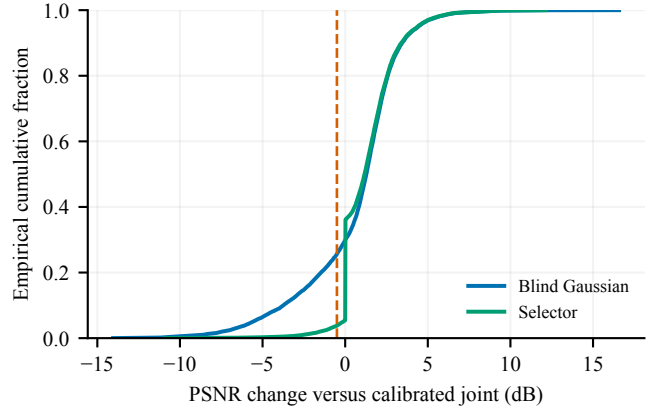


Fig. 2. Empirical CDF of per-frame PSNR change relative to calibrated joint sampling.

20.843 dB, so the thermal-noise-only port is insufficient as the sole Gaussian baseline.

To examine where selection helps, we compare fixed receivers across profiles. Under stationary interference, Gaussian denoising reaches 21.310 dB, compared with 19.282 dB for joint sampling. Under burst interference, joint sampling reaches 21.317 dB, compared with 20.481 dB for Gaussian denoising. The selector reaches 21.317, 21.285, and 22.177 dB for stationary, alternating, and burst interference, respectively. The reversed ordering of the fixed receivers supports adapting the disturbance model. It does not establish a causal interpretation of either energy statistic.

For external context, matched direct-codec anchors reach 17.394 dB for DeepJSCC and 16.923 dB for MambaJSCC. They use independently trained latent spaces and serve only as external codec references. One training seed and a fixed 40-epoch budget do not establish convergence or a general architecture ranking. Without fine-tuning, public DeepJSCC (ImageNet/SNR 19), MambaJSCC (CLIC2021/AWGN 10), and MambaJSCC (DIV2K/Rayleigh) weights yield 13.755, 17.020, and 16.650 dB, respectively. Their protocol mismatches preclude a controlled claim about retraining benefit. Figure 1 uses solid curves for the frozen SwinJSCC latent space and dashed curves for the separately trained codecs.

C. Inference Cost and Harmful Decisions

Figure 2 shows the frame-level CDF; the vertical line marks the -0.5 dB harmful-loss threshold. We next assess savings and harmful decisions. The selector chooses Gaussian reception on 69.5% of confirmation frames. It averages 52.2 predictor evaluations and 701 ms, versus 80 evaluations and 1180 ms for joint sampling. This saving does not remove harmful decisions. Against joint sampling, 3.79% of frames lose more than 0.5 dB, and the fifth percentile of the paired change is -0.127 dB (Fig. 2). The unavailable per-frame oracle reaches 21.760 dB; selection leaves 0.167 dB mean oracle regret. Mean MSE changes from 0.01849 to 0.01645, LPIPS from 0.3618 to 0.3362, and MS-SSIM from 0.8015 to 0.8318.

TABLE II
MEAN PSNR UNDER BLOCK MISMATCH AND NO INTERFERENCE.

Profile	Direct	Gaussian	Joint	Select
Shifted blocks	20.83	22.99	21.09	23.00
Random lengths	20.64	23.42	22.74	23.98
No interference	31.37	31.37	27.73	31.37

These mean improvements coexist with frame-level losses; selection provides no per-frame guarantee.

D. Block Mismatch and No Interference

To test block-partition sensitivity, we use 64 additional pairs, seed 3700, and SINR $-4, 4, 10$ dB. One profile shifts burst boundaries by half a receiver block; another draws segment lengths uniformly from 16 to 128 symbols and normalizes the binary on/off amplitude pattern. A separate SNR-20 condition has no interference. The receiver keeps its original block partition and frozen rule. We report these diagnostic families separately from matched confirmation. Table II reports the robustness tests. The selector changes PSNR relative to Gaussian reception by $+0.010$ dB for shifted blocks and $+0.563$ dB for random lengths. With no interference, it matches Gaussian reception and avoids the joint receiver's loss. The near-zero shifted-block gain limits transfer beyond the fitted partition. Diagnostics neither alter the rule nor enter confirmation averages. For a qualitative check, Fig. 3 shows all six prespecified confirmation examples: image pairs 512 and 527 under conditions 0, 8, and 15. No example was selected by reconstruction quality.

V. CONCLUSION

In this paper, we presented a blind selector that uses received energy and block-energy heterogeneity to choose between Gaussian denoising and joint signal–interference reconstruction with frozen neural weights. On independent confirmation, it improves PSNR by 0.750 dB over Gaussian denoising and 1.334 dB over joint sampling. It also uses 34.7% fewer predictor evaluations than joint sampling. Choosing the disturbance model per observation thus improves this frozen receiver stack without neural adaptation. The benefit remains population specific: harmful decisions persist, and shifted blocks nearly remove the advantage over Gaussian denoising. Fading, synchronization errors, and vehicular task accuracy require further evaluation.

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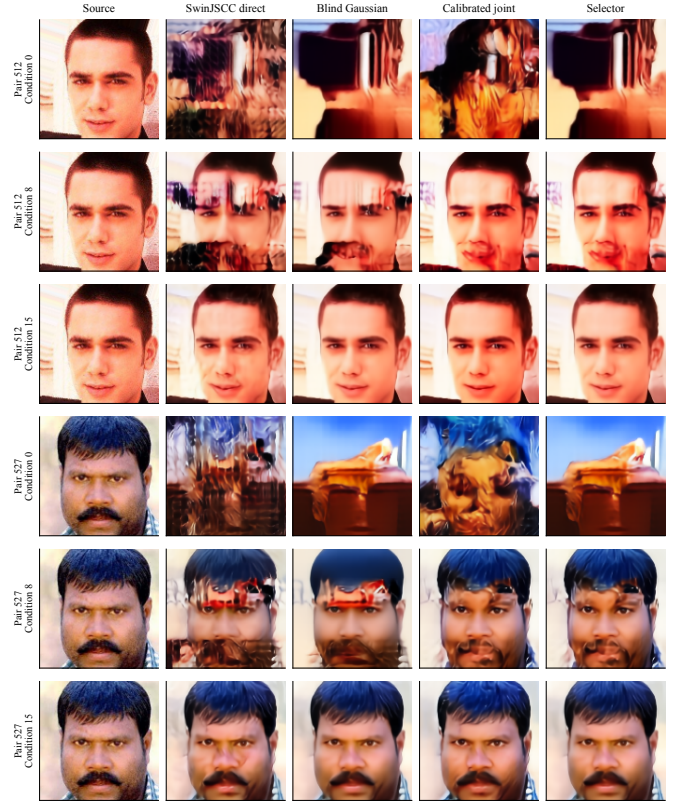


Fig. 3. Reconstructed images for six prespecified confirmation examples.

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